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# Inducing Parking Spaces to the Hyperoctahedral Group

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**ABSTRACT**

The permutation module $\mathrm{Park}_n$ of $\mathfrak{S}_n$ acting on parking functions of length $n$ appears in the decomposition of the space of diagonal coinvariants. This paper concerns the induction of the classical parking space to the hyperoctahedral group $B_n$, dubbed $\mathrm{SPark}_n$. We describe a set of generalized parking functions whose permutation module over $B_n$ coincides with this induction. Additionally, we introduce a generalization of Kreweras numbers giving the expansion of the Frobenius character of $\mathrm{SPark}_n$ in the $h_\mu(\mathbf{x})h_\nu(\mathbf{y})$ basis.

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## 1 Introduction

While studying the Hilbert scheme of points in the plane, Haiman [7] proved a character formula for the space $D$ of diagonal coinvariants (first conjectured in [5]). This character formula can be described in terms of certain combinatorial objects known as parking functions. A parking function is a tuple $\mathbf{a} = (a_1, a_2, \ldots, a_n) \in [n]^n$ whose order statistics satisfy $a_{(i)} \leq i$ for all $i \in [n] = \{1, 2, \ldots, n\}$. For any $n \in \mathbb{N}$, let $\mathcal{PF}_n$ denote the set of parking functions of length $n$. Parking functions were first studied by Pyke [14] and by

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Konheim and Weiss [10]; the latter also offered a combinatorial interpretation that justifies their "parking", which we describe next. Suppose $\mathbf{a}$ represents the parking preferences of $n$ cars parking on a one-way street with $n$ parking spots, labeled $1, 2, \ldots, n$. One at a time, the cars attempt to park in their preferred spots. If a car finds its preferred spot to be occupied, the car continues driving down the one-way street until it finds an empty spot in which to park. If it finds no such spot, the car leaves the street without parking. One can show that all cars park if and only if $\mathbf{a}$ is a parking function.

By definition, the set of parking functions is closed under permutation of the entries of a tuple (i.e., this action does not change its order statistics). Let $\mathrm{Park}_n$ denote the permutation module of $\mathfrak{S}_n$ acting on parking functions of length $n$. This space is related to the space $D$ of diagonal coinvariants studied by Haiman via

$$D \cong \mathrm{Park}_n \otimes \varepsilon,$$

where $\varepsilon$ is the sign representation of $\mathfrak{S}_n$.

In this paper, we are concerned with the induction of $\mathrm{Park}_n$ from $\mathfrak{S}_n$ to the hyperoctahedral group $B_n$. As a first attempt, this induction can be modeled as parking functions whose tuple entries are decorated with positive or negative signs, with no bearing on their combinatorial interpretation. Actually, these "naïve signed parking functions" coincide with the naïve shifted parking functions introduced by Stanley [17] when he studied a shiftification $\mathrm{SH}_n$ of the parking function symmetric function. We introduce a model of signed parking functions with richer combinatorics. While we find these signed parking functions interesting in their own right, we are also inspired in part by Stanley's search for a definition of shifted parking functions that more closely models the expansion of $\mathrm{SH}_n$ into certain bases. We quantify a connection between our signed parking functions and these shifted parking functions in Theorem 4.5.

In our model, the sign decorations dictate the direction of travel of each arriving car: cars with a positive sign traverse the street from left-to-right (traversing the parking spots in increasing order), whereas cars with a negative sign traverse the street from right-to-left (traversing the parking spots in decreasing order). A tuple $\mathbf{a} = (a_1, a_2, \ldots, a_n) \in \{-n, -(n-1), \ldots, -1, 1, 2, \ldots, n\}^n$ is a *signed parking function* if all cars park given the preferences in $\mathbf{a}$. Note that neither of these objects nor the naïve signed parking functions coincide with type $B$ parking functions, which are maximal chains in the type $B$ noncrossing partition lattice (refer to [16, Section 5]).

Our contributions in this paper both span and integrate the combinatorics of signed parking functions and the representation theory of the aforementioned induction. In Section 2, we provide some background on the representation theory of the hyperoctahedral group. In Section 3, we formally define the set $\mathcal{SPF}_n$ of signed parking functions of length $n$ and introduce involutions, called toggle maps, that allow us to both enumerate $\mathcal{SPF}_n$ and explain how it serves as a combinatorial model for $\mathrm{SPark}_n$, the induction of the classical parking space $\mathrm{Park}_n$ to the hyperoctahedral group $B_n$. In Section 4, we establish a decomposition of $\mathrm{SPark}_n$ that results in a generalization of Kreweras numbers that enumerate sets of generalized Dyck paths. We conclude with Section 5, where we state some open problems and directions for further study.

# 2 Background

We assume familiarity with the representation theory of the symmetric group and symmetric function notation (refer to [15] for a reference on both topics). Recall that the *Frobenius characteristic map* is an isomorphism between the ring $R_S$ of characters of symmetric group representations and the ring $\Lambda$ of symmetric functions. Given an $\mathfrak{S}_n$-representation $V$, the map is given by

$$\mathrm{ch}(V) = \frac{1}{n!} \sum_{\sigma \in \mathfrak{S}_n} \chi_V(\sigma) p_{\lambda(\sigma)},$$

where $\chi_V(\sigma)$ is the trace of $\sigma$ acting on $V$, $\lambda(\sigma)$ is the cycle type of $\sigma$, and $p_\lambda$ is the power-sum symmetric function indexed by $\lambda$.

## 2.1 Parking Spaces

The symmetric group $\mathfrak{S}_n$ acts on $\mathcal{PF}_n$ by permuting the entries of a preference tuple. We write $\mathrm{Park}_n$ for the corresponding permutation representation called the *parking space*. As previously mentioned, the parking space appears in formulas for the decomposition of the space of diagonal coinvariants (refer to [7]). The Frobenius characteristic of $\mathrm{Park}_n$ expands in the complete-homogeneous symmetric functions as

$$\mathrm{ch}(\mathrm{Park}_n) = \sum_{\lambda \vdash n} \mathrm{Krew}(\lambda) h_\lambda,$$

where $\lambda \vdash n$ denotes that $\lambda$ is a partition of $n$, $\mathrm{Krew}(\lambda)$ is the *Kreweras number* indexed by $\lambda$, and $h_\lambda$ is the complete-homogeneous symmetric function indexed by $\lambda$.

## 2.2 Representation Theory of the Hyperoctahedral Group

The hyperoctahedral group $B_n$ is the group of signed permutations. That is, maps $\eta$ from the set $\langle n \rangle \coloneqq \{-n, -n+1, \ldots, -2, -1, 1, 2, \ldots, n\}$ to itself with the property that

$$\eta(-i) = -\eta(i)$$

for all $i \in \langle n \rangle$. The group $B_n$ is generated by elements $s_i$ swapping $i$ and $i+1$ (and hence also swapping $-i$ and $-i-1$) for $i \in [n-1]$ and $t_i$ flipping the sign on $i$ for $i \in [n]$.

Important for our purposes, the group $B_n$ is isomorphic to a semidirect product

$$B_n \cong (\mathbb{Z}/2\mathbb{Z})^n \rtimes \mathfrak{S}_n.$$

In this way, every element of $B_n$ can be written in the form

$$\left( \prod_{i \in \Sigma} t_i \right) \pi$$

for some $\Sigma \subseteq [n]$ and $\pi \in \mathfrak{S}_n$ and $\pi t_i = t_{\pi(i)} \pi$ for all $i \in [n]$ and $\pi \in \mathfrak{S}_n$.

In this paper, we concern ourselves primarily with the induction of an $\mathfrak{S}_n$-representation to a $B_n$-representation. Recall that for $H$ a subgroup of $G$, the *induction* of an $H$-representation to a $G$-representation is given by

$$\mathrm{Ind}_H^G(V) = \mathbb{C}G \otimes_{\mathbb{C}H} V.$$

The structure of $B_n$ as a semidirect product leads immediately to the following lemma.

**Lemma 2.1.** *For $V$ an $\mathfrak{S}_n$-representation with basis indexed by $\mathcal{V}$, the induction*

$$\mathrm{Ind}_{\mathfrak{S}_n}^{B_n}(V) \cong \mathbb{C}\langle t_i : i \in [n]\rangle \otimes_{\mathbb{C}\mathfrak{S}_n} V$$

*has a basis indexed by $2^{[n]} \times \mathcal{V}$.*

To describe the irreducible characters of $B_n$-representations, we first recall that they are in bijection with conjugacy classes in $B_n$. Helpful toward this end will be writing an element $w$ of $B_n$ as a product of disjoint cycles. If a cycle $c = (x_1, x_2, \ldots, x_l)$ appears this way, then the cycle $-c = (-x_1, -x_2, \ldots, -x_l)$ also appears. If $c = -c$, we say that $c$ is a negative cycle of length $\frac{l}{2}$. If $c$ and $-c$ are distinct, then we call the product $(c)(-c)$ a positive cycle of length $l$. The signed cycle type of $w \in B_n$ is a pair $(\alpha, \beta)$ of integer partitions with $|\alpha| + |\beta| = n$ where $\alpha$ records the lengths of the positive cycles and $\beta$ records the lengths of the negative cycles.

**Example 2.2.** For all $i \in [7]$, let $\bar{i} = -i$. Let $w = 1\,\bar{2}\,\bar{4}\,7\,\bar{3}\,\bar{6}\,5 \in B_7$, which is the element sending

$$\begin{aligned} 1 &\mapsto \phantom{-}1,\\ 2 &\mapsto -2,\\ 3 &\mapsto -4,\\ 4 &\mapsto \phantom{-}7,\\ 5 &\mapsto -3,\\ 6 &\mapsto -6, \text{ and}\\ 7 &\mapsto \phantom{-}5. \end{aligned}$$

In disjoint cycle notation, $w$ can be written as

$$(1)(-1)(2,-2)(3,-4,-7,-5)(-3,4,7,5)(6,-6).$$

From this notation, we read off the signed cycle type of $w$ as the pair $((4,1),(1,1))$.

A detailed description of how the irreducible characters of $B_n$ are constructed can be found in [6]. The irreducible character $\chi^{(\lambda,0)} : B_n \to \mathbb{C}$ for $\lambda \vdash n$ is obtained by pulling back the character $\chi^\lambda : \mathfrak{S}_n \to \mathbb{C}$ across the surjection $B_n \to \mathfrak{S}_n$ that removes minus signs. Let $\delta : B_n \to \mathbb{C}$ be the character determined by $\delta(\tau_i) = -1$ for $i \in [n]$. That is, $\delta(w)$ records the parity of the number of positive numbers sent to a negative number by $w$. The remaining irreducible characters are obtained via the following two operations:

1. $\delta \otimes \chi^{(\mu,\nu)} = \chi^{(\nu,\mu)}$, and
2. $\chi^{(\mu,\nu)} = \mathrm{Ind}_{B_m \times B_n}^{B_{m+n}} \chi^{(\mu,0)} \cdot \chi^{(0,\nu)}$.

An analogous Frobenius characteristic map exists as an isomorphism between the ring $R_B$ of characters of $B_n$-representations and $\Lambda \otimes \Lambda$, which we think of as symmetric functions in two sets of variables $\mathbf{x}$ and $\mathbf{y}$, sending $\chi^{(\mu,\nu)}$ to $s_\mu(\mathbf{x})s_\nu(\mathbf{y})$. The maps $\mathrm{ch} : R_S \to \Lambda$ and $\mathrm{ch} : R_B \to \Lambda \otimes \Lambda$, allow us to consider induction as a map $\mathrm{Ind}_{\mathfrak{S}_n}^{B_n} : \Lambda \to \Lambda \otimes \Lambda$. The following lemma will be essential when inducing the classical parking space to the hyperoctahedral group. The result is given as the sole Example in [12, Appendix B].

**Lemma 2.3.** *For $\lambda \vdash n$,*

$$\mathrm{Ind}_{\mathfrak{S}_n}^{B_n}(\chi^\lambda) = \sum_{|\mu|+|\nu|=n} c_{\mu,\nu}^\lambda \chi^{(\mu,\nu)}$$

*where $c_{\mu,\nu}^\lambda$ are the Littlewood–Richardson coefficients.*

The analogous result for the complete homogeneous basis $h_\lambda$ follows from Example 25(a) of [12, Section 1.5] and multiplicativity of the coproduct on symmetric functions.

**Proposition 2.4.** *Let $\lambda \vdash n$. Then,*

$$\mathrm{Ind}_{\mathfrak{S}_n}^{B_n}(h_\lambda) = \sum_{\alpha+\beta=\lambda} h_\alpha(\mathbf{x})h_\beta(\mathbf{y})$$

*where the sum is over weak compositions $\alpha$ and $\beta$ summing to $\lambda$.*

# 3 Signed Parking Functions

In the classical parking problem posed by Konheim and Weiss [10], there is a sequence of $n$ cars parking on a one-way street with $n$ parking spots. Each car $i \in [n]$ has a preferred spot $a_i \in [n]$, to which it drives upon its arrival. If the spot is unoccupied, the car parks there. Otherwise, the car displaces down the one-way street and parks in the first unoccupied spot it encounters, if any. If there is no such spot, the car leaves the street unable to park. The cars' preferences are encoded in a tuple $\mathbf{a} = (a_1, a_2, \ldots, a_n) \in [n]^n$. If all cars are able to park, then $\mathbf{a}$ is said to be a *parking function* of length $n$. Let $\mathcal{PF}_n \subseteq [n]^n$ denote the set of parking functions of length $n \geq 1$. Konheim and Weiss [10] showed that $|\mathcal{PF}_n| = (n+1)^{n-1}$. It is also well-known that, if $\mathbf{a} = (a_1, a_2, \ldots, a_n) \in [n]^n$ and $\beta = (b_1, b_2, \ldots, b_n)$ is its weakly increasing rearrangement, then $\mathbf{a} \in \mathcal{PF}_n$ if and only if $1 \leq b_i \leq i$ for all $i \in [n]$. Refer to Martínez Mori [13] for an accessible introduction.

We introduce a new generalization of parking functions we term *signed parking functions*. In this setting, we consider signed preference tuples

$$\mathbf{a} = (a_1, a_2, \ldots, a_n) \in \langle n \rangle^n.$$

The interpretation is that there is a two-way street with $n$ parking spots, and in which no U-turns are allowed. There is a sequence of $n$ cars, each with a signed preference: the sign indicates whether the car enters the street from the left $(+)$ or from the right $(-)$. More formally, the signed parking rule is as follows:

**Definition 3.1** (Signed Parking Rule)**.** Upon its arrival, car $i \in [n]$ with preference $a_i \in \langle n \rangle$ drives to its preferred spot $|a_i|$. If the spot is unoccupied, the car parks there. Otherwise, there are two possibilities:

- If $a_i > 0$, then the car displaces down the street left-to-right. It probes the parking spots in increasing order and parks in the first unoccupied spot it encounters, if any.
- If $a_i < 0$, then the car displaces down the street right-to-left. It probes the parking spots in decreasing order and parks in the first unoccupied spot it encounters, if any.

If, using the signed parking rule, the preference tuple $\mathbf{a}$ allows all cars to park, then $\mathbf{a}$ is said to be a *signed parking function* of length $n$. Let $\mathcal{SPF}_n$ denote the set of signed parking functions of length $n \geq 1$.

Parking functions are signed parking functions, i.e. $\mathcal{PF}_n \subseteq \mathcal{SPF}_n$ for all $n \geq 1$. Now consider $\alpha = (1, -3, -3)$, which is not a parking function since its weakly increasing rearrangement $\beta = (b_1, b_2, b_3) = (-3, -3, 1)$ does not satisfy the inequality condition. For example, $1 \not\leq b_2 = -3$. However, $(1, -3, -3) \in \mathcal{SPF}_3 \setminus \mathcal{PF}_3$, i.e., it is a signed parking function since car 1 parks in spot 1, car 2 parks in spot 3, and car 3 parks in spot 2 after failing to park in spot 3 and traversing the street from right-to-left. However, $(1, 2, -1) \notin \mathcal{SPF}_3$ since car 3 is unable to park. In much the same way that permutations (in one-line notation) are parking functions, signed permutations are signed parking functions.

We now define some features of signed parking functions that are instrumental later in this work:

- For each $\emptyset \subseteq \Sigma \subseteq [n]$, let

$$\mathcal{SPF}_n(\Sigma) \coloneqq \{\mathbf{a} = (a_1, a_2, \ldots, a_n) \in \mathcal{SPF}_n : a_i < 0 \iff i \in \Sigma, \forall i \in [n]\}.$$

  For any $\mathbf{a} \in \mathcal{SPF}_n(\Sigma)$, we say $\Sigma$ is its *signature*. For example, let $\mathbf{a} = (1, -3, -3) \in \mathcal{SPF}_3$. Its signature is $\Sigma = \{2, 3\}$, whereas the signature of any (classical) parking function is $\Sigma = \emptyset$. Note that

$$\mathcal{SPF}_n = \bigsqcup_{\emptyset \subseteq \Sigma \subseteq [n]} \mathcal{SPF}_n(\Sigma)$$

  for each $n \geq 1$, and where $\bigsqcup$ denotes a disjoint union.

- For any $\mathbf{a} = (a_1, a_2, \ldots, a_n) \in \mathcal{SPF}_n$, its *outcome*

$$\mathcal{O}(\mathbf{a}) = (\mathcal{O}(\mathbf{a})_1, \mathcal{O}(\mathbf{a})_2, \ldots, \mathcal{O}(\mathbf{a})_n) \in \mathfrak{S}_n$$

  is the permutation in one-line notation that encodes which car parks in which spot. In particular, spot $i \in [n]$ is occupied by car $\mathcal{O}(\mathbf{a})_i \in [n]$. For example, the outcome of $\mathbf{a} = (1, -4, -2, 2) \in \mathcal{SPF}_4$ is $\mathcal{O}(\mathbf{a}) = (1, 3, 4, 2) \in \mathfrak{S}_4$, in one-line notation. The inverse $\mathcal{O}(\mathbf{a})^{-1} = (\mathcal{O}(\mathbf{a})_1^{-1}, \mathcal{O}(\mathbf{a})_2^{-1}, \ldots, \mathcal{O}(\mathbf{a})_n^{-1})$, tells us that for each $i \in [n]$, car $i$ parked in spot $\mathcal{O}(\mathbf{a})_i^{-1}$. For example, if again $\mathbf{a} = (1, -4, -2, 2)$, then the inverse of $\mathcal{O}(\mathbf{a}) = (1, 3, 4, 2) \in \mathfrak{S}_4$ is $\mathcal{O}(\mathbf{a})^{-1} = (1, 4, 2, 3) \in \mathfrak{S}_4$, in one-line notation.

- The *displacement* of car $i \in [n]$ is the distance between its preferred spot and the spot in which it ultimately parks. More formally, the displacement of car $i \in [n]$ under $\mathbf{a} = (a_1, a_2, \ldots, a_n) \in \mathcal{SPF}_n(\Sigma)$ is

$$D_{\mathbf{a}}(i) \coloneqq \begin{cases} \mathcal{O}(\mathbf{a})_i^{-1} - a_i & \text{if } i \notin \Sigma, \\ |a_i| - \mathcal{O}(\mathbf{a})_i^{-1} & \text{if } i \in \Sigma. \end{cases}$$

The *total displacement* of $\mathbf{a} \in \mathcal{SPF}_n$ is $D_{\mathbf{a}} = \sum_{i=1}^{n} D_{\mathbf{a}}(i)$. For example, the total displacement of $\mathbf{a} = (1, -4, -2, 2) \in \mathcal{SPF}_4$ is

$$D_{\mathbf{a}} = (1-1) + (4-4) + (2-2) + (3-2) = 1.$$

## 3.1 Toggles

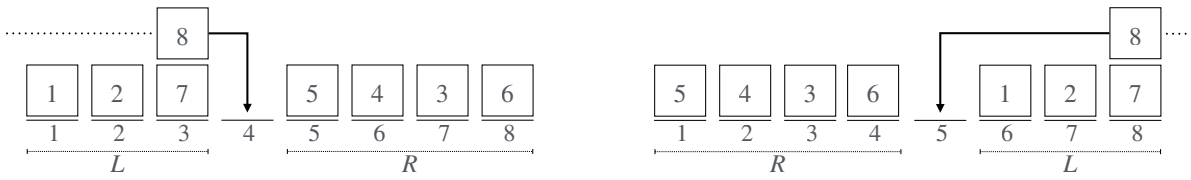
In this section we define a set of functions $\{\tau_i\}_{i\in[n]}$ called *toggles*. For each $i \in [n]$, the function $\tau_i : \mathcal{SPF}_n \to \mathcal{SPF}_n$ is a toggle in that, if $\mathbf{a} \in \mathcal{SPF}_n(\Sigma)$ for some $\emptyset \subseteq \Sigma \subseteq [n]$, then $\tau_i(\mathbf{a}) \in \mathcal{SPF}_n(\Sigma')$ where

$$\Sigma' = \begin{cases} \Sigma \setminus \{i\} & \text{if } i \in \Sigma, \\ \Sigma \cup \{i\} & \text{if } i \notin \Sigma. \end{cases}$$

In other words, a toggle maps a signed parking function with a given signature to a signed parking function whose signature differs at the toggled index. As we elaborate on below, these maps are generally not just a change of sign at the toggled index.

Before developing our general technical definition of the toggles, we motivate it through our definition of $\tau_n$. Let $\mathbf{a} \in \mathcal{SPF}_n$ and consider the street upon the arrival of car $n$. Since $\mathbf{a} \in \mathcal{SPF}_n$, there is exactly one unoccupied spot $i \in [n]$. Let $L, R \subseteq [n-1]$ denote the subsets of cars parked to the left and right of spot $i$, respectively. Since spot $i$ is unoccupied, the parking processes of $L$ and $R$ are independent. Therefore, we define $\tau_n$ to uniformly shift the preferences of the cars in $L$ so that they park on the rightmost segment of the street. Similarly, we define $\tau_n$ to uniformly shift the preferences of the cars in $R$ so that they park on the leftmost segment of the street. This shifts the index of the unoccupied spot from $i$ to $n - i + 1 \in [n]$. Therefore, we also define $\tau_n$ to adapt the preference of car $n$ through a shift and change of sign so that it traverses the same set of cars under $\tau_n(\mathbf{a})$ as it does under $\mathbf{a}$. We illustrate this in the following example.

**Example 3.2.** Let $\mathbf{a} = (1, 1, 7, -7, 5, 5, 3, 3) \in \mathcal{SPF}_8(\Sigma)$, where $\Sigma = \{4\}$. As depicted in Figure 1(A), its outcome is $\mathcal{O}(\mathbf{a}) = (1, 2, 7, 8, 5, 4, 3, 6)$. Upon the arrival of car 8, spot 4 is empty. The cars parked to the left and right of spot 4 are $L = \{1, 2, 7\}$ and $R = \{5, 4, 3, 6\}$, respectively. As shown by the dashed line and arrow, car 8 with preference 3 traverses cars 1, 2, and 7 before parking in spot 4.




(A) $\mathcal{O}(\mathbf{a}) = (1, 2, 7, 8, 5, 4, 3, 6)$. (B) $\mathcal{O}(\tau_n(\mathbf{a})) = (5, 4, 3, 6, 8, 1, 2, 7)$.

Figure 1: An application of the toggle $\tau_8$ on $\mathbf{a} = (1, 1, 7, -7, 5, 5, 3, 3) \in \mathcal{SPF}_8(\{4\})$ results in $\tau_8(\mathbf{a}) = (6, 6, 3, -3, 1, 1, 8, -8) \in \mathcal{SPF}_8(\{4, 8\})$. The outcomes of $\mathbf{a}$ and $\tau_8(\mathbf{a})$ upon the arrival of car 8 are illustrated in Figures 1(A) and 1(B), respectively.

To have the cars in $L$ park on the rightmost segment of the street, we shift each of their preferences 5 spots to the right, i.e., we increase the magnitude of $a_j$ by 5 for each $j \in L$.

To have the cars in $R$ park on the leftmost segment of the street, we shift each of their preferences 4 spots to the left, i.e., we decrease the magnitude of $a_j$ by 4 for each $j \in R$. Lastly, to have car 8 traverse the same set of cars, we shift its preference 5 spots to the right and change its sign, i.e., we increase the magnitude of $a_8$ by 5 and change its sign from positive to negative. The resulting preference is

$$\begin{aligned}\tau_8(\mathbf{a}) &= (1+5, 1+5, 7-4, -(7-4), 5-4, 5-4, 3+5, -(3+5)) \\ &= (6, 6, 3, -3, 1, 1, 8, -8) \in \mathcal{SPF}_8(\Sigma'),\end{aligned}$$

where $\Sigma' = \{4, 8\}$. As depicted in Figure 1(B), its outcome is $\mathcal{O}(\tau_8(\mathbf{a})) = (5, 4, 3, 6, 8, 1, 2, 7)$.

For ease of exposition going forward, we introduce the following notation. Given $a, b \in \mathbb{Z}$, with $a \neq 0$, let

$$a \oplus b := \begin{cases} a + b & \text{if } a > 0, \\ a - b & \text{if } a < 0. \end{cases}$$

We now formally define $\tau_n$.

**Definition 3.3.** Let $\mathbf{a} = (a_1, a_2, \ldots, a_n) \in \mathcal{SPF}_n$, $i \in [n]$ be the unoccupied spot upon the arrival of car $n$, and $L, R \subseteq [n-1]$ be the subsets of cars parked to the left and right of spot $i$, respectively. The toggle of $\mathbf{a}$ at $n$ is $\tau_n(\mathbf{a}) = (\tau_n(\mathbf{a})_1, \tau_n(\mathbf{a})_2, \ldots, \tau_n(\mathbf{a})_n)$, where

$$\tau_n(\mathbf{a})_j = \begin{cases} a_j \oplus (n-i+1) & \text{if } j \in L, \\ a_j \oplus (-i) & \text{if } j \in R, \\ -(a_n \oplus (n-i+1)) & \text{if } j = n \text{ and } |a_n| \in [i-1], \\ -(a_n \oplus (-i)) & \text{if } j = n \text{ and } |a_n| \in [n] \setminus [i], \\ -\operatorname{sgn}(a_n)(n-i+1) & \text{if } j = n \text{ and } |a_n| = i. \end{cases}$$

for all $j \in [n]$.

We illustrate Theorem 3.3 using the previous example.

**Example 3.4** (Theorem 3.2 continued)**.** If $\mathbf{a} = (1, 1, 7, -7, 5, 5, 3, 3) \in \mathcal{SPF}_8(\{4\})$, then $L = \{1, 2, 7\}$, $R = \{5, 4, 3, 6\}$, and $i = 4$. For $j \in L$, by Theorem 3.3 we have that

- $\tau_8(\mathbf{a})_1 = a_1 \oplus (8-4+1) = 1 \oplus 5 = 6$,
- $\tau_8(\mathbf{a})_2 = a_2 \oplus (8-4+1) = 1 \oplus 5 = 6$, and
- $\tau_8(\mathbf{a})_7 = a_7 \oplus (8-4+1) = 3 \oplus 5 = 8$.

For $j \in R$, by Theorem 3.3, we have that

- $\tau_8(\mathbf{a})_3 = a_3 \oplus (-4) = 7 \oplus (-4) = 3$,
- $\tau_8(\mathbf{a})_4 = a_4 \oplus (-4) = -7 \oplus (-4) = -3$,
- $\tau_8(\mathbf{a})_5 = a_5 \oplus (-4) = 5 \oplus (-4) = 1$, and
- $\tau_8(\mathbf{a})_6 = a_6 \oplus (-4) = 5 \oplus (-4) = 1$.

Since $a_8 = 3 \in [i-1] = \{1,2,3\}$ and its signature is positive, we have that $\tau_8(\mathbf{a})_8 = -(a_8 \oplus (8-4+1)) = -(a_8+8-4+1) = -8$. Putting this together we have that $\tau_8(\mathbf{a}) = (6,6,3,-3,1,1,8,-8) \in \mathcal{SPF}_8(\{4,8\})$, which agrees with the conclusion in Theorem 3.2.

Given any $k \in [n]$ and $\mathbf{a} \in \mathcal{SPF}_n$, consider the state of the street immediately after the $k$th car parks under the preferences in $\mathbf{a}$. Let $S_k(\mathbf{a}) \subseteq [k]$ be the largest set of cars contiguously parked on the segment of the street containing the spot in which the $k$th car parked. We now define the general toggle map $\tau_k$ on the elements in $\mathcal{SPF}_n$.

**Definition 3.5.** Given a signature $\Sigma$, let $\Sigma'$ be the result of toggling whether $k$ is in $\Sigma$. Given a signed parking function $\mathbf{a} \in \mathcal{SPF}_n(\Sigma)$, let $\tau_k(\mathbf{a}) \in \mathcal{SPF}_n(\Sigma')$ be the signed parking function defined as follows. Consider the segment $S = S_k(\mathbf{a})$. This subsection of the street can then be treated as its own street, and by shifting the preferences of cars parking in $S$ so that the smallest preference is one, we can consider the cars parking in this segment as their own signed parking function $\beta \in \mathcal{SPF}_m$ where $m$ denotes the length of the street segment $S$. We then modify the preferences of the cars parking in $S$ by applying $\tau_m$ to $\beta$ as in Theorem 3.3 and then shift the preferences back so that the cars park in the original segment $S$. Note that because car $k$ corresponds to the last car to park in $\beta$, we have flipped the sign on the $k$th preference.

We illustrate Theorem 3.5 next.

**Example 3.6.** Let $\mathbf{a} = (1,5,2,-5,2,-6,-8,-8) \in \mathcal{SPF}_8(\{4,6,7,8\})$. Suppose $k = 4$; the state of the street upon the arrival of car 4 is illustrated in Figure 2(A). To compute




(A) (B)

Figure 2: Figure 2(A) depicts parking the first four cars using the preference list $\mathbf{a} = (1,5,2,-5,2,-6,-8,-8)$. Figure 2(B) depicts parking the first four cars using the preference list $\tau_4(\mathbf{a}) = (1,4,2,4,2,-6,-8,-8)$.

$\tau_4(1,5,2,-5,2,-6,-8,-8)$, first note that $S_4(\mathbf{a}) = \{2,4\}$. The left and right segments of the sub-street corresponding to $S_4(\mathbf{a})$ are $L = \emptyset$ and $R = \{2\}$, respectively. Toggling at 4 decreases the preferences of the car in $R$ by 1. Since car 4 originally traversed the cars in $R$, we adjust its preference to spot 4 so that it traverses the same set of cars after the toggle. This gives us that $\tau_4(1,5,2,-5,2,-6,-8,-8) = (1,4,2,4,2,-6,-8,-8)$, as illustrated in Figure 2(B). Under the signed parking process, $\tau_4(\mathbf{a}) = (1,4,2,4,2,-6,-8,-8)$ yields the outcome $(1,3,5,2,4,6,8,7)$.

**Proposition 3.7.** *The toggle maps $\tau_k$ are involutions.*

*Proof.* The contiguous segment $S = S_k(\mathbf{a})$ containing car $k$ just after car $k$ parks remains unchanged upon applying $\tau_k$, hence it suffices to show that $\tau_n$ is an involution on $\mathcal{SPF}_n$. Let $L$ and $R$ be as in Theorem 3.3 for the application of $\tau_n$ to $\mathbf{a}$. Let $L'$ and $R'$ be the analogous sets for the application of $\tau_n$ to $\tau_n(\mathbf{a})$. Note that the toggle exchanges whether

a car parks to the left or the right of the $n$th car. That is, $L' = R$ and $R' = L$. If the spot remaining unoccupied upon the arrival of car $n$ in $\mathbf{a}$ is spot $i$, then the spot remaining unoccupied upon the arrival of car $n$ in $\tau_n(\mathbf{a})$ is spot $i' = n - i + 1$.

Suppose $j \in L$. Then

$$\tau_n(\tau_n(\mathbf{a}))_j = (a_j \oplus (n - i + 1)) \oplus (-i') = a_j.$$

If instead $j \in R$, then

$$\tau_n(\tau_n(\mathbf{a}))_j = (a_j \oplus (-i)) \oplus (n - i' + 1) = a_j.$$

Finally, suppose $j = n$. If car $n$ had preferred a spot to the left of where it parked, then $a_n \in [i - 1]$, and

$$\tau_n(\mathbf{a})_n = -(a_n \oplus (n - i + 1)).$$

Then in $\tau_n(\mathbf{a})$, car $n$ prefers a spot to the right of where it parked, so

$$\tau_n(\tau_n(\mathbf{a}))_n = -(-(a_n \oplus (n - i + 1)) \oplus (-i')) = a_n.$$

The case that car $n$ preferred the spot where it parked or to the right of where it parked proceeds similarly. □

We now give a count for the number of signed parking functions.

**Theorem 3.8.** *For $n \geq 1$,*

$$|\mathcal{SPF}_n| = 2^n |\mathcal{PF}_n| = 2^n (n+1)^{n-1},$$

*which corresponds to OEIS sequence A097629.*

*Proof.* By Theorem 3.7, the sets $\{\mathcal{SPF}_n(\Sigma) : \Sigma \subseteq [n]\}$ are equinumerous. Moreover, they are each of size $|\mathcal{SPF}_n(\emptyset)| = |\mathcal{PF}_n|$. Therefore, $|\mathcal{SPF}_n| = 2^n |\mathcal{PF}_n|$. □

Next, we give an alternative proof of Theorem 3.8. While this proof is much more involved than our first proof, it sets the stage for our proof of a result in Section 4.

*Alternative Proof of Theorem 3.8.* Consider a sequence of $n$ cars parking on a circular two-way street with $n + 1$ parking spots, numbered $1, 2, \ldots, n, n + 1$. Cars with positive preferences traverse the street clockwise, whereas cars with negative preferences traverse the street counter-clockwise. Let

$$\mathcal{SPP}_n := \{\mathbf{a} = (a_1, a_2, \ldots, a_n) \in \langle n + 1 \rangle^n\}$$

be the set of all possible signed parking preference tuples for $n$ cars. Note that $\mathcal{SPF}_n \subseteq \mathcal{SPP}_n$ and that $|\mathcal{SPP}_n| = (2(n + 1))^n$.

For any $\mathbf{a} = (a_1, a_2, \ldots, a_n) \in \mathcal{SPP}_n$ and $s \in \mathbb{N}_0 = \mathbb{N} \cup \{0\}$, let

$$\pi^{(s)}(\mathbf{a}) = (r_1^{(s)}(\mathbf{a}), r_2^{(s)}(\mathbf{a}), \ldots, r_n^{(s)}(\mathbf{a})) \in \mathcal{SPP}_n,$$

where

$$r_i^{(s)}(\mathbf{a}) := \begin{cases} n+1, & \text{if } a_i > 0 \text{ and } a_i + s \mod (n+1) = 0; \\ a_i + s \mod (n+1) & \text{if } a_i > 0 \text{ and } a_i + s \mod (n+1) \neq 0; \\ -(n+1), & \text{if } a_i < 0 \text{ and } a_i - s \mod (n+1) = 0; \text{ and} \\ -(a_i - s \mod (n+1)), & \text{otherwise,} \end{cases}$$

for all $i \in [n]$. Note that $|\{\pi^{(s)}(\mathbf{a}) : s \in \mathbb{N}_0\}| = n+1$. Since the choice of $\mathbf{a} \in \mathcal{SPP}_n$ was arbitrary, $\pi^{(s)}$ partitions $\mathcal{SPP}_n$ into equivalence classes, each of size $n+1$.

Now, fix any $\mathbf{a} \in \mathcal{SPP}_n$ and consider the sequence of $n$ cars parking on the circular two-way street with $n+1$ numbered parking spots under any signed parking preference tuple $\mathbf{a}' \in \{\pi^{(s)}(\mathbf{a}) : s \in \mathbb{N}_0\}$. Since there are $n+1$ spots but only $n$ cars, all cars ultimately park and leave exactly one unoccupied spot. Note that exactly one $\mathbf{a}' \in \{\pi^{(s)}(\mathbf{a}) : s \in \mathbb{N}_0\}$ results in spot $n+1$ ultimately left unoccupied: the successive elements in $\{\pi^{(s)}(\mathbf{a}) : s \in \mathbb{N}_0\}$ rotate the unoccupied spot clockwise around the circle one spot at a time. Therefore, there are

$$\frac{1}{n+1}(2(n+1))^n = 2^n(n+1)^{n-1}$$

parking preferences that result in spot $n+1$ ultimately left unoccupied. These preferences form the set $\mathcal{SPF}_n$, which establishes the count $|\mathcal{SPF}_n| = 2^n(n+1)^{n-1}$. □

In Theorem 3.10 we prove that the toggles commute. Before giving the proof we illustrate the result with an example.

**Example 3.9** (Continuing Theorem 3.6)**.** Let

$$\mathbf{a} = (1, 5, 2, -5, 2, -6, -8, -8) \in \mathcal{SPF}_8(\{4, 6, 7, 8\}),$$

which has parking outcome $\mathcal{O}(\mathbf{a}) = (1, 3, 5, 4, 2, 6, 8, 7)$.
To compute $\tau_5(1, 5, 2, -5, 2, -6, -8, -8)$ we note $S_5(\mathbf{a}) = \{1, 3, 5, 4, 2\}$ and the left and right segments of the substreet $S_5(\mathbf{a})$ are $L = \{1, 3\}$ and $R = \{4, 2\}$, respectively. Toggling at 5 takes the preferences of the cars in $L$ and shifts them up by 3, so they now park to the right of where car 5 parks. Denote this set of cars now parking to the right by $R'$. Similarly, the toggle takes the preferences of the cars in $R$ and lowers them by 3, so they now park to the left of where car 5 parks. Denote this set of cars now parking to the left by $L'$. As car 5 hit $L$, it must now hit $R'$ at the same place, hence the preference for car 5 is changed to $-5$. This gives that $\tau_5(\mathbf{a}) = (4, 2, 5, -2, -5, -6, -8, -8)$. Parking the cars with $\tau_5(\mathbf{a}) = (4, 2, 5, -2, -5, -6, -8, -8)$ yields the outcome $(4, 2, 5, 1, 3, 6, 8, 7)$. Recall from Theorem 3.6 that $\tau_4(\mathbf{a}) = (1, 4, 2, 4, 2, -6, -8, -8)$.

To compute $\tau_4(\tau_5(\mathbf{a}))$ we note $S_4(\tau_5(\mathbf{a})) = \{4, 2\}$ with left segment $L = \emptyset$ and right segment $R = \{2\}$. Toggling at 4 takes the preference of the car in $R$ and lowers it by 1. As car 4 hit $R$, it must now hit $L'$ at the same place, hence the preference for car 4 is changed to 1. All other preferences remain unchanged. This gives

$$\tau_4(\tau_5(\mathbf{a})) = (4, 1, 5, 1, -5, -6, -8, -8).$$

To compute $\tau_5(\tau_4(\mathbf{a}))$ we note $S_5(\tau_4(\mathbf{a})) = \{1, 3, 5, 2, 4\}$ with left segment $L = \{1, 3\}$ and right segment $R = \{2, 4\}$. Toggling at 5 takes the preference of the cars in $L$ and shifts

them up by 3, and takes the preferences of the cars in $R$ and lowers them by 3. As car 5 hit $L$, it must now hit $R'$ at the same place, hence the preference for car 5 is changed to $-5$. All other preferences remain unchanged. This gives that

$$\tau_5(\tau_4(\mathbf{a})) = (4, 1, 5, 1, -5, -6, -8, -8).$$

We conclude by observing that $\tau_4(\tau_5(\mathbf{a})) = (4, 1, 5, 1, -5, -6, -8, -8) = \tau_5(\tau_4(\mathbf{a}))$.

## 3.2 As a model for the induced representation

In addition to their enumerative utility, the toggle maps $\tau_i$ allow us to define a hyperoctahedral action on the set $\mathcal{SPF}_n$ in which $t_i$ acts as $\tau_i$. In the following lemma, we first establish that the toggle maps satisfy the relations among the elements $t_i$.

**Lemma 3.10.** *The toggle maps on $\mathcal{SPF}_n$ satisfy the relations $\tau_i\tau_j = \tau_j\tau_i$ for all $i, j \in [n]$.*

*Proof.* The case when $i = j$ is immediate from Theorem 3.7, so we may assume without loss of generality that $i < j$. When cars $i$ and $j$ lie in disjoint connected components on the street upon parking the cars $[j]$, the two toggles affect disjoint sets of preferences and hence commute. Suppose then that, upon parking the cars $[j]$ the cars $i$ and $j$ lie in the same connected component on the street. We prove the case when $i$ parks to the left of $j$ and both cars have positive preference, as the other cases proceed similarly. See Figure 3 for an illustration of the resulting outcomes upon applying the toggles in either order. Let

- $L$ be the set of cars to the left of car $i$ in its connected component after cars $[i]$ park,
- $R$ be the set of cars to the right of car $i$ in its connected component after cars $[i]$ park, and
- $S$ be the set of cars to the right of car $j$ in its connected component after cars $[j]$ park.

For example, the sizes of these sets in Figure 3 are $|L| = |R| = 2$ and $|S| = 5$. We then need only verify that the preference of each car affected by both toggles is transformed in the same way upon applying the toggles in either order.

For a car $x \in L$, applying $\tau_j\tau_i$ replaces the preference $\mathbf{a}_x$ with

$$(\mathbf{a}_x \oplus (|R| + 1)) \oplus (|S| + 1)$$

and applying $\tau_i\tau_j$ yields

$$(\mathbf{a}_x \oplus (|S| + 1)) \oplus (|R| + 1).$$

Both of these values equal $\mathbf{a}_x \oplus (|R| + |S| + 2)$. We summarize this computation and those for cars $b \in R$ and $c \in S$ as well as cars $i$ and $j$ in the following five diagrams where $\ell = |L|$, $r = |R|$, and $s = |S|$. Horizontal arrows correspond to applying $\tau_i$ and vertical arrows correspond to applying $\tau_j$:

$$\begin{array}{ccc} \mathbf{a}_x & \xrightarrow{\oplus(r+1)} & \mathbf{a}_x \oplus (r+1) \\ {\scriptstyle\oplus(s+1)}\downarrow & & \downarrow{\scriptstyle\oplus(s+1)} \\ \mathbf{a}_x \oplus (s+1) & \xrightarrow[\oplus(r+1)]{} & \mathbf{a}_x \oplus (r+s+2) \end{array} \qquad \begin{array}{ccc} \mathbf{a}_y & \xrightarrow{\oplus(-\ell-1)} & \mathbf{a}_y \oplus (-\ell-1) \\ {\scriptstyle\oplus(s+1)}\downarrow & & \downarrow{\scriptstyle\oplus(s+1)} \\ \mathbf{a}_y \oplus (s+1) & \xrightarrow[\oplus(-\ell-1)]{} & \mathbf{a}_y \oplus (s-\ell) \end{array} \qquad \begin{array}{ccc} \mathbf{a}_z & \longrightarrow & \mathbf{a}_z \\ {\scriptstyle\oplus(s+1)}\downarrow & & \downarrow{\scriptstyle\oplus(s+1)} \\ \mathbf{a}_z \oplus (s+1) & \longrightarrow & \mathbf{a}_z \oplus (s+1) \end{array}$$

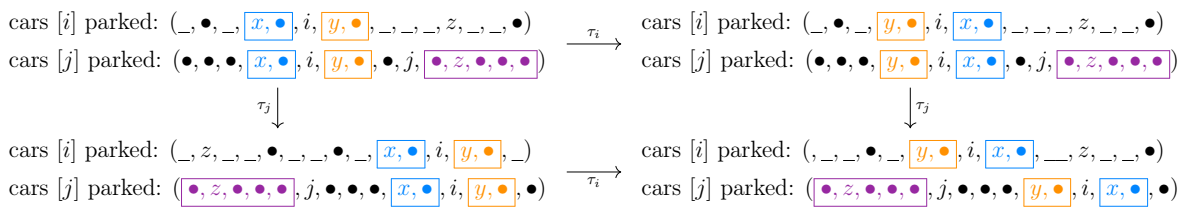



Figure 3: An illustration of the change in outcomes upon applying toggles $\tau_i$ and $\tau_j$ in both orders. We display the partial outcomes upon parking the first $i$ cars, then the first $j$ cars for $i < j$. To save space, we show only the connected component containing $i$ and $j$ upon parking the first $j$ cars. Namely, the boxed entries to the left of $i$ (or $j$) represent the left and right contiguous segments as in Theorem 3.5.

$$\begin{array}{ccc} \mathbf{a}_i & \longrightarrow & -\mathbf{a}_i \oplus (2D_{\mathbf{a}}(i)) \\ {\scriptstyle \oplus(s+1)}\downarrow & & \downarrow{\scriptstyle \oplus(s+1)} \\ \mathbf{a}_i \oplus (s+1) & \longrightarrow & -\mathbf{a}_i \oplus (2D_{\mathbf{a}}(i) + s + 1) \end{array}$$

where $D_{\mathbf{a}}(i)$ is the displacement of car $i$.

$$\begin{array}{ccc} \mathbf{a}_j & \longrightarrow & \mathbf{a}_j \\ \downarrow & & \downarrow \\ -\mathbf{a}_j \oplus (2D_{\mathbf{a}}(j)) & \longrightarrow & -\mathbf{a}_j \oplus (2D_{\mathbf{a}}(j)) \end{array}$$

where $D_{\mathbf{a}}(j)$ is the displacement of car $j$.

□

Now for $\Sigma \subseteq [n]$, we define

$$\pi_\Sigma = \prod_{i \in \Sigma} \tau_i.$$

Note that by Theorem 3.10, this product can be taken in any order. We now define a map $\sigma_i : \mathcal{SPF}_n \to \mathcal{SPF}_n$ as follows. For $\mathbf{a} \in \mathcal{SPF}_n(\Sigma)$, we first obtain $\pi_\Sigma(\mathbf{a}) \in \mathcal{PF}_n$. We then allow the simple transposition $s_i$ to act by transposing the preferences of cars $i$ and $i+1$, then map by $\pi_{s_i(\Sigma)}$, where we note that

$$s_i(\Sigma) = \begin{cases} \Sigma & \text{if } i, i+1 \in \Sigma \text{ or if } i, i+1 \notin \Sigma \\ (\Sigma \setminus \{i\}) \cup \{i+1\} & \text{if } i \in \Sigma \text{ and } i+1 \notin \Sigma \\ (\Sigma \setminus \{i+1\}) \cup \{i\} & \text{if } i+1 \in \Sigma \text{ and } i \notin \Sigma. \end{cases}$$

Hence, we define

$$\sigma_i(\mathbf{a}) = \pi_{s_i(\Sigma)} s_i \pi_\Sigma(\mathbf{a}).$$

Next, we show that the maps $\sigma_i$ and $\tau_i$ satisfy the relations between $s_i$ and $t_i$ in $B_n$.

**Lemma 3.11.** *For $i \in [n]$, we have that $\sigma_i \tau_n = \tau_{s_i(n)} \sigma_i$.*

*Proof.* First note that because each $\tau_i$ is an involution, we have that

$$\pi_{\Sigma \Delta \{i\}} = \pi_\Sigma \tau_i$$

where $\Sigma\Delta\{i\}$ denotes the symmetric difference of $\Sigma$ and $\{i\}$.

Let $\mathbf{a} \in \mathcal{SPF}_n(\Sigma)$. Then $\tau_n(\mathbf{a}) \in \mathcal{SPF}_n(\Sigma\Delta\{n\})$, so

$$\begin{aligned}\sigma_i(\tau_n(\mathbf{a})) &= \pi_{s_i(\Sigma\Delta\{n\})} \circ s_i \circ \pi_{\Sigma\Delta\{n\}} \circ \tau_n(\mathbf{a}) \\ &= \pi_{s_i(\Sigma)\Delta\{s_i(n)\}} \circ s_i \circ \pi_\Sigma(\mathbf{a}) \\ &= \tau_{s_i(n)} \circ \pi_{s_i(\Sigma)} \circ s_i \circ \pi_\Sigma(\mathbf{a}) \\ &= \tau_{s_i(n)}(\sigma_i(\mathbf{a})).\end{aligned}$$

□

**Theorem 3.12.** *As $B_n$-representations, the permutation module $\mathbb{C}\mathcal{SPF}_n$ where $s_i$ acts as $\sigma_i$ and $t_i$ acts as $\tau_i$ is isomorphic to* $\mathrm{SPark}_n = \mathrm{Ind}_{\mathfrak{S}_n}^{B_n}(\mathrm{Park}_n)$.

*Proof.* Define a map $\varphi : \mathrm{SPark}_n \to \mathbb{C}\mathcal{SPF}_n$ by

$$\left(\prod_{i\in\Sigma} t_i\right) \otimes \mathbf{a} \mapsto \pi_\Sigma(\mathbf{a}).$$

By Theorem 3.8, $\varphi$ is an isomorphism of vector spaces. By Theorem 3.10 and Theorem 3.11, $\varphi$ is a map of $B_n$-representations. □

# 4 Signed Kreweras Numbers

In this section, we concern ourselves with a particular decomposition of the space $\mathrm{SPark}_n$ obtained by writing its Frobenius characteristic in the $h_\mu(\mathbf{x})h_\nu(\mathbf{y})$ basis.

To begin, we recall that for a partition $\lambda \vdash n$ with $k$ parks, the Kreweras number is defined [11] as

$$\mathrm{Krew}(\lambda) = \frac{1}{n+1}\binom{n+1}{n-k, m_1(\lambda), m_2(\lambda), \ldots, m_n(\lambda)},$$

where $m_i(\lambda)$ is the multiplicity of $i$ in $\lambda$.

**Definition 4.1.** Given $\mu \vdash a$ and $\nu \vdash b$ with $a+b = n$, define the *signed Kreweras number* $\mathrm{Krew}(\mu, \nu)$ to be

$$\mathrm{Krew}(\mu, \nu) = \sum_{\alpha,\beta} \mathrm{Krew}(\alpha + \beta),$$

where the sum is over weak compositions $\alpha$ and $\beta$ each with $\ell(\mu) + \ell(\nu)$ parts such that $\mathrm{sort}(\alpha)$ with zeros removed is equal to $\mu$, $\mathrm{sort}(\beta)$ with zeros removed is equal to $\nu$, and $\alpha + \beta$ with zeros removed (without sorting) is a partition of $n$ in which case we write $(\alpha + \beta) \vdash n$.

**Example 4.2.** The partitions obtained (with multiplicity) by adding permutaistions of $\mu = (2, 2)$ and $\nu = (2, 1)$ as described in Theorem 4.1 so that $\alpha + \beta$ with zeros removed

is a partition of 7 are

$$\begin{aligned}
(2,2,0,0)+(0,0,2,1)&=(2,2,2,1),\\
(2,0,2,0)+(0,2,0,1)&=(2,2,2,1),\\
(0,2,2,0)+(2,0,0,1)&=(2,2,2,1),\\
(2,2,0,0)+(2,0,1,0)&=(4,2,1),\\
(2,0,2,0)+(1,2,0,0)&=(3,2,2),\\
(2,2,0,0)+(1,0,2,0)&=(3,2,2),\ \text{and}\\
(2,2,0,0)+(2,1,0,0)&=(4,3).
\end{aligned}$$

Note, for example, $(2,2,0,0)+(1,2,0,0)=(3,4,0,0)$ is not considered as $(3,4)$ is not a partition of 7 since the parts are not in weakly decreasing order.

Hence, we compute the corresponding signed Kreweras number as

$$\begin{aligned}
\operatorname{Krew}((2,2),(2,1)) &= 3\operatorname{Krew}((2,2,2,1))+\operatorname{Krew}((4,2,1))+2\operatorname{Krew}((3,2,2))+\operatorname{Krew}((4,3))\\
&= 3\cdot\frac{1}{8}\binom{8}{4,1,3}+\frac{1}{8}\binom{8}{5,1,1,1}+2\cdot\frac{1}{8}\binom{8}{5,2,1}+\frac{1}{8}\binom{8}{6,1,1}\\
&= 3\cdot 35+42+2\cdot 21+7\\
&= 196.
\end{aligned}$$

**Remark 4.3.** For $\mu$ and $\nu$ integer partitions, we have immediately from the definition of signed Kreweras numbers that

1. $\operatorname{Krew}(\mu,0)=\operatorname{Krew}(\mu)$, and
2. $\operatorname{Krew}(\mu,\nu)=\operatorname{Krew}(\nu,\mu)$.

**Theorem 4.4.** *The Frobenius character of the signed parking space* $\mathrm{SPark}_n$ *is given by*

$$\operatorname{ch}(\mathrm{SPark}_n)=\sum_{\substack{\mu,\nu\\ |\mu|+|\nu|=n}}\operatorname{Krew}(\mu,\nu)h_\mu(\mathbf{x})h_\nu(\mathbf{y}).$$

*Proof.* It follows that

$$\begin{aligned}
\operatorname{Ind}_{\mathfrak{S}_n}^{B_n}(\operatorname{ch}(\mathrm{Park}_n)) &= \sum_{\lambda\vdash n}\operatorname{Krew}(\lambda)\operatorname{Ind}_{\mathfrak{S}_n}^{B_n}(h_\lambda)\\
&= \sum_{\lambda\vdash n}\sum_{\alpha+\beta=\lambda}\operatorname{Krew}(\lambda)h_\alpha(\mathbf{x})h_\beta(\mathbf{y})\\
&= \sum_{(\alpha+\beta)\vdash n}\operatorname{Krew}(\alpha+\beta)h_\alpha(\mathbf{x})h_\beta(\mathbf{y}) \qquad (1)\\
&= \sum_{\substack{\mu,\nu\\ |\mu|+|\nu|=n}}\operatorname{Krew}(\mu,\nu)h_\mu(\mathbf{x})h_\nu(\mathbf{y}),
\end{aligned}$$

where in Equation (1) we recall that $(\alpha+\beta)\vdash n$ denotes that $\alpha+\beta$ with zeros removed is a partition of $n$. □

Before moving on to a combinatorial description of the signed Kreweras numbers, we close this section by pointing out a close connection with the shiftification $\mathrm{SH}_n$ of the parking function symmetric function introduced by Stanley [17].

**Theorem 4.5.** *The Frobenius character of the restriction of* $\mathrm{SPark}_n$ *to the symmetric group* $\mathfrak{S}_n$ *expands in the power-sum basis as*

$$\mathrm{Res}^{B_n}_{\mathfrak{S}_n}(\mathrm{ch}(\mathrm{SPark}_n)) = \sum_{\lambda \vdash n} z_\lambda{}^{-1} 2^{\ell(\lambda)} (n+1)^{\ell(\lambda)-1} p_\lambda.$$

*Proof.* To compute the character of this permutation module, we need only compute the number of signed parking functions fixed by any permutation $\sigma \in \mathfrak{S}_n$, the set of which we denote by $\mathrm{SPark}_n^\sigma$. More precisely, the coefficient on $p_\lambda$ is $z_\lambda{}^{-1}|\mathrm{SPark}_n^\sigma|$ where $\sigma$ is any permutation with cycle type $\lambda$.

Let $\sigma$ have cycle type $\lambda$. Note that if a signed parking function $\mathbf{a}$ has signature $\Sigma$, the signed parking function $\sigma.\mathbf{a}$ has signature $\sigma(\Sigma)$. We have that $\sigma(\Sigma) = \Sigma$ if and only if any cycle of $\sigma$ that intersects with $\Sigma$ is entirely contained in $\Sigma$. There are $2^{\ell(\lambda)}$ ways to choose a subset of the cycles of $\sigma$ to comprise $\Sigma$, hence there are $2^{\ell(\lambda)}$ possible signatures among elements of $\mathrm{SPark}_n^\sigma$.

Fixing a signature $\Sigma$ as described above, we now need only determine how many elements of $\mathrm{SPark}_n^\sigma$ have this signature. Recall that the set of signed parking functions $\mathbf{a}$ with signature $\Sigma$ is in bijection with classical parking functions $\mathbf{b}$ where $\mathbf{a} = \pi_\Sigma(\mathbf{b})$. The condition that $\sigma(\Sigma) = \Sigma$ guarantees that

$$\begin{aligned} \sigma.\mathbf{a} &= \sigma.\pi_\Sigma(\mathbf{b}) \\ &= \pi_\Sigma(\sigma.\mathbf{b}). \end{aligned}$$

We then have that $\sigma.\mathbf{a} = \mathbf{a}$ is equivalent to $\sigma.\mathbf{b} = \mathbf{b}$. The number of classical parking functions fixed by $\sigma$ is $(n+1)^{\ell(\lambda)-1}$, hence

$$|\mathrm{SPark}_n^\sigma| = 2^{\ell(\lambda)}(n+1)^{\ell(\lambda)-1},$$

and we have proven the desired expansion. □

**Remark 4.6.** Comparing the expansion in Theorem 4.5 to [17, Theorem 2.1], we note that Stanley's shiftification $\mathrm{SH}_n$ of the parking function symmetric function is obtained by deleting each term from $\mathrm{Res}^{B_n}_{\mathfrak{S}_n}(\mathrm{ch}(\mathrm{SPark}_n))$ indexed by a partition with an even-sized part. In terms of the shiftification operator defined in [17], the shiftification of $\mathrm{Res}^{B_n}_{\mathfrak{S}_n}(\mathrm{ch}(\mathrm{SPark}_n))$ then coincides with the shiftification of $\mathrm{ch}(\mathrm{Park}_n)$.

## 4.1 Signed Dyck Paths

In this section, we introduce signed Dyck paths in order to give a combinatorial interpretation of signed Kreweras numbers.

**Definition 4.7.** A *signed Dyck path* is a Dyck path in which each up-step is labeled with a sign $+$ or $-$, subject to the condition that no up-step labeled $-$ immediately follows one labeled $+$. Let $\mathrm{SD}_n$ be the set of all signed Dyck paths from $(0,0)$ to $(n,n)$ and let $\mathrm{SD}(\mu,\nu)$ be the set of signed Dyck paths whose runs of positive and negative up-steps are given by $\mu$ and $\nu$ respectively. We recall that a run of up-steps is a maximal consecutive sequence of up-steps.

**Example 4.8.** The illustration in Figure 4 shows a signed Dyck path in $\mathrm{SD}((2,1,1),(2,1)) \subseteq \mathrm{SD}_7$.

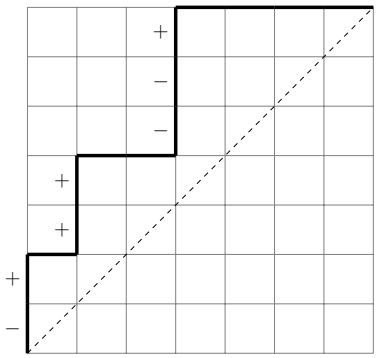



Figure 4: A signed Dyck path whose runs of up-steps have multiplicities of positive and negative steps given by $(1,2,1)$ and $(1,0,2)$ respectively. Sorting these tuples into partitions shows us that this signed Dyck path lies in $\mathrm{SD}((2,1,1),(2,1))$.

**Proposition 4.9.** *The number of signed Dyck paths with runs of positive and negative up-steps given by $\mu$ and $\nu$ respectively is*

$$|\mathrm{SD}(\mu,\nu)| = \mathrm{Krew}(\mu,\nu).$$

*Proof.* Let $D_n$ denote the set of (unlabeled) Dyck paths. We demonstrate the equality by providing a bijection between $\mathrm{SD}_n$ and the set of triples $(\alpha,\beta,p)$ where $\alpha$ and $\beta$ are weak compositions, $p \in D_n$, and the sum $\alpha+\beta$ is the partition recording the runs of up-steps of $p$. The result then follows from the fact that $\mathrm{Krew}(\lambda)$ counts the number of $p \in D_n$ with runs of up-steps given by $\lambda$.

Let $q \in \mathrm{SD}_n$ and for each run of up-steps, we have a tuple $(a,b)$ recording the number of positive and negative up-steps in the run respectively. Arrange these tuples in decreasing order of their sum, leaving tuples with identical sums in the same relative order. For example, the signed Dyck path in Theorem 4.8 would produce the tuples $(1,2),(1,1),(2,0)$. Take the first entries of these tuples as the weak composition $\alpha$ and the second entries as the weak composition $\beta$. Then let $p$ be the result of removing the $+/-$ labels from $q$. This process produces the desired triple $(\alpha,\beta,p)$. Given any such triple, we can use the compositions $\alpha$ and $\beta$ to recover where the $+/-$ labels should appear on the Dyck path. □

**Proposition 4.10.** *The total number of signed Dyck paths is*

$$|\mathrm{SD}_n| = \frac{1}{n+1}\binom{3n+1}{n}.$$

*Proof.* First we count the total number of lattice paths from $(0,0)$ to $(n,n)$ with up-steps labeled as in Theorem 4.7. Given such a path $p$, we can form an unlabeled lattice path $q$ from $(0,0)$ to $(2n+1,n)$ as follows. For each positive up-step at position $i$ of $p$, the path $q$ has an up-step at position $2i$. For each negative up-step at position $i$ of $p$, the path $q$ has an up-step at position $2i+1$ (see Figure 5).

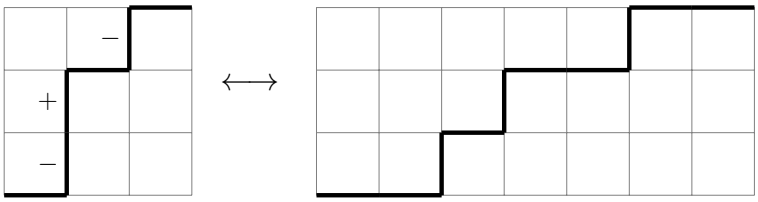



Figure 5: A signed lattice path and its corresponding unlabeled lattice path.

Given a signed lattice path from $(0,0)$ to $(n,n)$, we associate a word consisting of $n$ letters $E$ and $n$ letters $+$ and $-$ in some order. For example, the signed lattice path in Figure 5 corresponds to the word

$$E-+E-E.$$

We define the rotation of such a word $w$ as follows. Place an $E$ at the start of the word, then move the first letter to the end repeatedly until $E$ is again the first letter. Remove the initial $E$ to obtain $w'$, the rotation of $w$. For example, if $w = E-+E-E$, its rotation is $w' = -+E-EE$. By a classical argument*, each orbit under rotation has $n+1$ elements and exactly one element in each orbit is a Dyck path, hence

$$|\mathrm{SD}_n| = \frac{1}{n+1}\binom{3n+1}{n}. \qquad \square$$

We say a tuple $(a_1, a_2, \ldots, a_n) \in \mathbb{Z}^n$ is *weakly increasing* if $a_i \leq a_{i+1}$ for all $i \in [n-1]$. Let $\mathcal{SPF}_n^{\uparrow}$ denote the set of weakly increasing signed parking functions of length $n$. Next, we show that the number of weakly increasing signed parking functions is a Catalan relative. Our proof extends the technique in [9] to the signed parking function case. We illustrate the proof in Figure 6 and Theorem 4.12.

**Theorem 4.11.** *The number of weakly increasing signed parking functions of length $n \geq 1$ is*

$$|\mathcal{SPF}_n^{\uparrow}| = \frac{1}{n+1}\binom{3n+1}{n},$$

*which is OEIS sequence A006013.*

*Proof.* Consider a sequence of $n$ cars parking on a circular two-way street with $n+1$ parking spots, numbered $1, 2, \ldots, n, n+1$. Cars with positive preferences traverse the street clockwise, whereas cars with negative preferences traverse the street counter-clockwise. Let

$$\mathcal{SPP}_n^{\uparrow} \coloneqq \{\mathbf{a} = (a_1, a_2, \ldots, a_n) \in \langle n+1\rangle^n : a_1 \leq a_2 \leq \cdots \leq a_n\} \subseteq \langle n+1\rangle^n$$

be the set of all possible weakly increasing, signed parking preference tuples for $n$ cars. Note that $\mathcal{SPF}_n^{\uparrow} \subseteq \mathcal{SPP}_n^{\uparrow}$. A direct application of the balls and bars method shows that

$$|\mathcal{SPP}_n^{\uparrow}| = \left(\!\binom{2(n+1)}{n}\!\right) = \binom{2(n+1)+n-1}{n} = \binom{3n+1}{n}.$$

* This is a version of the cycle lemma given by Dvoretzky and Motzkin in [3].

For any weakly increasing, signed parking preference tuple $\mathbf{a} \in \mathcal{SPP}_n^{\uparrow}$, let

$$\kappa(\mathbf{a}) = (k_{-(n+1)}(\mathbf{a}), k_{-n}(\mathbf{a}), \ldots, k_{-2}(\mathbf{a}), k_{-1}(\mathbf{a}), k_1(\mathbf{a}), k_2(\mathbf{a}), \ldots, k_n(\mathbf{a}), k_{n+1}(\mathbf{a})) \in [n]_0^{2(n+1)}$$

be its content, where

$$k_j(\mathbf{a}) \coloneqq |\{i \in [n] : a_i = j\}|$$

for all $j \in \langle n+1 \rangle$. In other words, $k_j(\mathbf{a})$ is the multiplicity (or number of occurrences) of $j \in \langle n+1 \rangle$ in $\mathbf{a}$. Moreover, let $\kappa(\mathcal{SPP}_n^{\uparrow}) = \{\kappa(\mathbf{a}) : \mathbf{a} \in \mathcal{SPP}_n^{\uparrow}\} \subseteq [n]_0^{2(n+1)}$ be the set of contents of the set of weakly increasing, signed parking preference tuples for $n$ cars. Note that $|\kappa(\mathcal{SPP}_n^{\uparrow})| = |\mathcal{SPP}_n^{\uparrow}|$, as a weakly increasing, signed parking preference tuple is uniquely identified by its content.

Next, we define a *signed circular shift* operation $\omega : \kappa(\mathcal{SPP}_n^{\uparrow}) \to \kappa(\mathcal{SPP}_n^{\uparrow})$. For each element

$$k = (k_{-(n+1)}, k_{-n}, \ldots, k_{-2}, k_{-1}, k_1, k_2, \ldots, k_n, k_{n+1}) \in \kappa(\mathcal{SPP}_n^{\uparrow}),$$

let

$$\omega(k) \coloneqq (k_{-n}, \ldots, k_{-2}, k_{-1}, k_{-(n+1)}, k_{n+1}, k_1, k_2, \ldots, k_n) \in \kappa(\mathcal{SPP}_n^{\uparrow}).$$

In other words, $\omega$ applies a pair of mirrored circular shift operations:

- It applies a clockwise circular shift on the positively-indexed elements of $\kappa$. That is, it moves the rightmost positively-indexed entry to the leftmost positively-indexed position and shifts all other positively-indexed entries to the next position on the right.
- It applies a counter-clockwise circular shift on the negatively-indexed elements of $\kappa$. That is, it moves the leftmost negatively-indexed entry to the rightmost negatively-indexed position and shifts all other negatively-indexed entries to the next position on the left.

Any content $k = (k_{-(n+1)}, k_{-n}, \ldots, k_{-2}, k_{-1}, k_1, k_2, \ldots, k_n, k_{n+1}) \in \kappa(\mathcal{SPP}_n^{\uparrow})$ has $2(n+1)$ (nonnegative integer) entries and satisfies $\sum_{j \in [-(n+1),(n+1)]_{\hat{0}}} k_i = n$, so that there always exists at least one pair $j, j' \in [-(n+1), n+1]_{\hat{0}}$ with $k_j \neq 0$ and $k'_j = 0$. Therefore, $\omega$ partitions the elements of $\kappa(\mathcal{SPP}_n^{\uparrow})$ into equivalence classes each of size $n+1$; let $\mathcal{K}$ denote this collection of equivalence classes. For each equivalence class $K \in \mathcal{K}$, fix an arbitrary element as its canonical representative and denote it by $\hat{k} \in K$. Moreover, for each element $k \in K$, let $\zeta(k) = (z_1(k), z_2(k), \ldots, z_n(k)) \in \mathcal{SPP}_n^{\uparrow}$ be its corresponding weakly increasing, signed parking preference tuple. Note that the sets $\{\zeta(k) : k \in K\}$ for $K \in \mathcal{K}$, each of size $n+1$, partition the set $\mathcal{SPP}_n^{\uparrow}$.

Next, consider any equivalence class $K \in \mathcal{K}$ and its canonical element $\hat{k} \in K$. For any $s \in \mathbb{N}_0$, let $\rho^{(s)}(\hat{k}) = (r_1^{(s)}(\hat{k}), r_2^{(s)}(\hat{k}), \ldots, r_n^{(s)}(\hat{k})) \in \langle n+1 \rangle^n$ where

$$r_i^{(s)}(\hat{k}) \coloneqq \begin{cases} n+1, & \text{if } z_i(\hat{k}) > 0 \text{ and } z_i(\hat{k}) + s \mod (n+1) = 0; \\ z_i(\hat{k}) + s \mod (n+1) & \text{if } z_i(\hat{k}) > 0 \text{ and } z_i(\hat{k}) + s \mod (n+1) \neq 0; \\ -(n+1), & \text{if } z_i(\hat{k}) < 0 \text{ and } z_i(\hat{k}) - s \mod (n+1) = 0; \\ \text{and } -(z_i(\hat{k}) - s \mod (n+1)), & \text{otherwise,} \end{cases}$$

for all $i \in [n]$. Note that $\kappa(\rho^{(s)}(\hat{k}))$, i.e. the content of $\rho^{(s)}(\hat{k})$, is equivalently obtained by performing $s$ signed circular shift operations on $\hat{k}$, i.e. $\omega^{(s)}(\hat{k})$. Therefore, the sets $K$ and $\{\rho^{(s)}(\hat{k}) : s \in \mathbb{N}_0\}$ are in bijection. By transitivity, the sets $\{\zeta(k) : k \in K\}$ and $\{\rho^{(s)}(\hat{k}) : s \in \mathbb{N}_0\}$ are also in bijection, and each of them is of size $n+1$. Let $\phi : \{\rho^{(s)}(\hat{k}) : s \in \mathbb{N}_0\} \to \{\zeta(k) : k \in K\}$ be the bijective map given by $\phi(\mathbf{a}) = \zeta(\kappa(\mathbf{a}))$ for all $\mathbf{a} \in \{\rho^{(s)}(\hat{k}) : s \in \mathbb{N}_0\}$, i.e. $\phi = \zeta \circ \kappa$.

Now, consider the sequence of $n$ cars parking on the circular two-way street with $n+1$ numbered parking spots under any signed parking preference tuple $\mathbf{a} \in \{\rho^{(s)}(\hat{k}) : s \in \mathbb{N}_0\} \subseteq \langle n+1 \rangle^n$. Note that while $\mathbf{a}$ need not be weakly increasing, its negative entries precede its positive entries. Since there are $n+1$ spots but only $n$ cars, all cars ultimately park and leave exactly one unoccupied spot. We claim that this is the same spot ultimately left unoccupied if the cars arrive under the weakly increasing, signed preference tuple $\phi(\mathbf{a}) \in \mathcal{SPP}_n^{\uparrow}$. To prove this, we use a result in [4, Theorem 3.2] which shows that for the classical parking protocol, the set of ultimately occupied spots is invariant under rearrangement of the entries of the preference tuple, even when there are more spots than cars.

Let $0 \leq \ell \leq n$ be the number of negative entries in $\mathbf{a}$, all of which appear in its first $\ell$ positions. Then, $\ell$ is also the number of negative entries in $\phi(\mathbf{a})$, all of which appear in its first $\ell$ positions. The first $\ell$ positions of $\mathbf{a}$ and $\phi(\mathbf{a})$ are rearrangements of each other, and we treat them as classical (unsigned) parking functions except the cars traverse the street from right to left. By [4, Theorem 3.2], the set of spots ultimately occupied after the first $\ell$ cars arrive under $\mathbf{a}$ is the same set of spots ultimately occupied after the first $\ell$ cars arrive under $\phi(\mathbf{a})$. Similarly, the final $n-\ell$ positions of $\mathbf{a}$ and $\phi(\mathbf{a})$ are rearrangements of each other, and we treat them as classical (unsigned) parking functions in which the cars traverse the street from left to right. By [4, Theorem 3.2], the set of spots ultimately after the final $n-\ell$ cars arrive under $\mathbf{a}$ is the same set of spots ultimately after the final $n-\ell$ cars arrive under $\phi(\mathbf{a})$. This shows that the spot ultimately left unoccupied if the cars arrive under $\mathbf{a}$ is the same spot ultimately left unoccupied if the cars arrive under $\phi(\mathbf{a})$.

Lastly, note that exactly one $\mathbf{a} \in \{\rho^{(s)}(\hat{k}) : s \in \mathbb{N}_0\}$ results in spot $n+1$ ultimately left unoccupied: the successive elements in $\{\rho^{(s)}(\hat{k}) : s \in \mathbb{N}_0\}$ rotate the unoccupied spot clockwise around the circle one spot at a time. Therefore, by our previous claim, exactly one $\mathbf{a} \in \{\zeta(k) : k \in K\}$ results in spot $n+1$ ultimately left unoccupied. Since the sets $\{\zeta(k) : k \in K\}$ for $K \in \mathcal{K}$ are each of size $n+1$ and partition $\mathcal{SPP}_n^{\uparrow}$, which is of size $\binom{3n+1}{n}$, there are

$$\frac{1}{n+1}\binom{3n+1}{n}$$

weakly increasing parking preferences that result in spot $n+1$ ultimately left unoccupied. These preferences form the set $\mathcal{SPF}_n^{\uparrow}$, which establishes the count $|\mathcal{SPF}_n^{\uparrow}| = \frac{1}{n+1}\binom{3n+1}{n}$. □

**Example 4.12.** Consider 5 cars parking on a circular two-way street with 6 spots, i.e., after spot 6, the street loops back into spot 1. The set of weakly increasing signed preferences of length 5 are in bijection with the contents of length 12 via the content map $\kappa$ whose inverse is the weakly increasing word map $\zeta$. For example, $\kappa((-5,-4,5,5,6)) = (0,1,1,0,0,0;0,0,0,0,2,1)$ and $\zeta((0,1,1,0,0,0;0,0,0,0,2,1)) = (-5,-4,5,5,6)$.

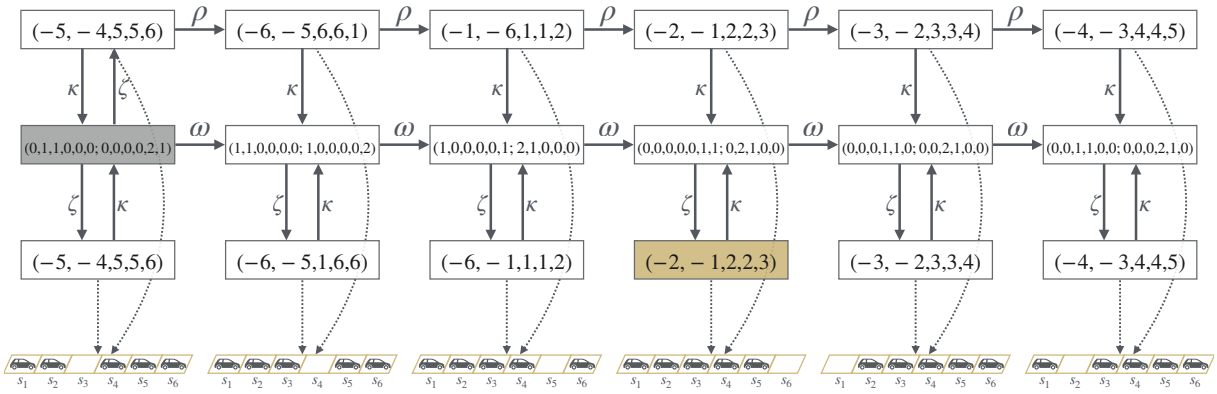



Figure 6: Illustrations accompanying the proof of Theorem 4.11 and Theorem 4.12.

The signed circular shift operation, denoted by $\omega$, partitions the set of contents into equivalence classes of size 6. In particular, the elements of the orbit

$$\begin{aligned}(0,1,1,0,0,0;0,0,0,0,2,1) &\xrightarrow{\omega} (1,1,0,0,0,0;1,0,0,0,0,2)\\ &\xrightarrow{\omega} (1,0,0,0,0,1;2,1,0,0,0,0)\\ &\xrightarrow{\omega} (0,0,0,0,1,1;0,2,1,0,0,0)\\ &\xrightarrow{\omega} (0,0,0,1,1,0;0,0,2,1,0,0)\\ &\xrightarrow{\omega} (0,0,1,1,0,0;0,0,0,2,1,0)\end{aligned}$$

are all in the same equivalence class: without loss of generality, let

$$\hat{k} = (0,1,1,0,0,0;0,0,0,0,2,1)$$

be its canonical representative (shaded in grey in Figure 6). We also rotate its corresponding weakly increasing signed preference $\zeta(\hat{k}) = (-5,-4,5,5,6)$ via the rotation map $\rho$, which increases all of its entries by one modulo $n+1$, as in Pollak's original proof. This forms the orbit

$$\begin{aligned}(-5,-4,5,5,6) &\xrightarrow{\rho} (-6,-5,6,6,1) \xrightarrow{\rho} (-1,-6,1,1,2)\\ &\xrightarrow{\rho} (-2,-1,2,2,3) \xrightarrow{\rho} (-3,-2,3,3,4) \xrightarrow{\rho} (-4,-3,4,4,5),\end{aligned}$$

also of size 6, and with exactly one (not necessarily weakly increasing) parking function: $(-2,-1,2,2,3)$. Through $\kappa$, these preference rotations are in bijection with the equivalence class of contents. For example,

$$\kappa\left((-1,-6,1,1,2)\right) = (1,0,0,0,0,1;2,1,0,0,0,0).$$

Moreover, the composition $\phi = \zeta \circ \kappa$ bijectively maps these preference rotations to weakly increasing preferences. For example,

$$\phi\left((-1,-6,1,1,2)\right) = \zeta\left((1,0,0,0,0,1;2,1,0,0,0,0)\right) = (-6,-1,1,1,2).$$

Finally, the spot ultimately left unoccupied is invariant under $\phi$, as noted by the dashed arrows and the unlabeled parking outcomes. For example, the spot ultimately left unoccupied under both $(-1,-6,1,1,2)$ and $(-6,-1,1,1,2)$ is spot 5. Since $(-2,-1,2,2,3)$ is the unique parking function in its equivalence class of preference rotations, $(-2,-1,2,2,3)$ is the unique parking function in its equivalence class of weakly increasing preferences (shaded in golden in Figure 6).

**Theorem 4.13.** *The number of $\mathfrak{S}_n$-orbits of $\mathcal{SPF}_n$ is given by*

$$\frac{1}{n+1}\binom{3n+1}{n}.$$

*Proof.* The number of orbits of $\mathfrak{S}_n$ acting on $\mathcal{SPF}_n$ is given by the multiplicity of the trivial representation in the restriction $\mathrm{Res}^{B_n}_{\mathfrak{S}_n}(\mathrm{SPark}_n)$. Using Frobenius reciprocity along with Theorem 2.4, we compute

$$\begin{aligned}\left\langle \mathrm{Res}^{B_n}_{\mathfrak{S}_n}(\mathrm{ch}(\mathrm{SPark}_n)), h_n \right\rangle &= \left\langle \mathrm{ch}(\mathrm{SPark}_n), \mathrm{Ind}^{B_n}_{\mathfrak{S}_n}(h_n) \right\rangle \\ &= \sum_{i+j=n} \left\langle \mathrm{ch}(\mathrm{SPark}_n), h_i(x)h_j(x) \right\rangle,\end{aligned}$$

where the sum is over pairs $(i,j)$ of nonnegative integers summing to $n$. Expanding $\mathrm{ch}(\mathrm{SPark}_n)$ using Theorem 4.4, we obtain

$$\sum_{i+j=n} \left\langle \mathrm{ch}(\mathrm{SPark}_n), h_i(x)h_j(x) \right\rangle = \sum_{\mu,\nu} \mathrm{Krew}(\mu,\nu) \sum_{i+j=n} \left\langle h_\mu(x)h_\nu(y), h_i(x)h_j(y) \right\rangle.$$

Because $\langle h_\mu, h_i \rangle = \begin{cases} 1 & \text{if } i = |\mu| \\ 0 & \text{if } i \neq |\mu| \end{cases}$, we have that

$$\left\langle \mathrm{Res}^{B_n}_{\mathfrak{S}_n}(\mathrm{ch}(\mathrm{SPark}_n)), h_n \right\rangle = \sum_{\mu,\nu} \mathrm{Krew}(\mu,\nu).$$

By Theorem 4.9 and Theorem 4.10, this sum of signed Kreweras numbers is equal to $\frac{1}{n+1}\binom{3n+1}{n}$. □

# 5 Open Problems and Future Work

In Theorem 4.10 we showed that the number of signed Dyck paths of length $n$ is $\frac{1}{n+1}\binom{3n+1}{n}$. This yields OEIS sequence A097629 which also counts the number of all unrooted directed trees on $n+1$ nodes, also known as signed trees on $[n+1]$. Ardila, Beck, and McWhirter [1, Proposition 5.1] establish that there are $2^n(n+1)^{n-1}$ signed trees on $[n+1]$. This leads us to pose the following.

**Problem 5.1.** Give a direct bijection between signed parking functions and signed trees.

Additionally, Theorem 4.11 shows that weakly increasing signed parking functions are also counted by $\frac{1}{n+1}\binom{3n+1}{n}$. Thus we pose the following.

**Problem 5.2.** Give a bijection between signed Dyck paths of length $n$ and weakly increasing signed parking functions of length $n$ where the area statistic is carried to the displacement statistic.

Classical parking functions have a beautiful characterization in terms of inequalities on the weakly-increasing rearrangement of a preference vector. We ask for an analogous characterization for signed parking functions.

**Problem 5.3.** Give a characterization of signed parking functions via inequalities analagous to that for classical parking functions.

There is also still much to understand about the action of a permutation in $\mathfrak{S}_n$ on $\mathcal{SPF}_n$. In particular, it would be helpful to have a purely combinatorial description of the action of a simple transposition $s_i$ on an element of $\mathcal{SPF}_n$ (rather than needing to apply toggle maps before and after).

Furthermore, from Theorem 4.11 and Theorem 4.13, we have that the number of weakly increasing signed parking functions and the number of $\mathfrak{S}_n$-orbits on $\mathcal{SPF}_n$ are equinumerous. This observation leads to the following conjecture.

**Conjecture 5.4.** Each $\mathfrak{S}_n$-orbit of $\mathcal{SPF}_n$ contains a unique weakly increasing signed parking function.

In [17], Stanley asks for a set of "shifted parking functions" to combinatorially explain the decomposition of a symmetric function $\mathrm{SH}_n$ into the basis $V_\lambda$ where $\lambda \vdash n$ has only odd parts. The symmetric function $\mathrm{SH}_n$ is closely related to the signed parking functions (see Theorem 4.5). Although Stanley's question is resolved in [8], we are curious whether signed parking functions, which are defined purely from a parking process, could also describe this decomposition. In order for signed parking functions to serve Stanley's purposes, one would need to resolve the following question.

**Problem 5.5.** Partition the set $\mathcal{SPF}_n$ into blocks as follows. For each partition $\lambda \vdash n$ into *odd parts*, there are $a_\lambda$ blocks of size $d_\lambda$ where $a_\lambda$ is the coefficient of $V_\lambda$ in $\mathrm{SH}_n$ and

$$d_\lambda = 2^{n-\ell(\lambda)} \binom{n}{\lambda_1, \lambda_2, \ldots}.$$

In addition, one can consider the following. We say a finite sequence $(a_1, a_2, \ldots, a_n) \in \mathbb{Z}^n$ is *absolutely weakly increasing* if $|a_i| \leq |a_{i+1}|$ for all $i \in [n-1]$. Let $\mathcal{SPF}_n^{|\uparrow|}$ denote the set of absolutely weakly increasing signed parking functions of length $n$.

**Problem 5.6.** Prove that the number of absolutely weakly increasing signed parking functions of length $n \geq 1$ is

$$|\mathcal{SPF}_n^{|\uparrow|}| = 2^n C_n,$$

where $C_n = \frac{1}{n+1}\binom{2n}{n}$ denotes the $n$th Catalan number, and this sequence corresponds to OEIS sequence A000108.

The above problem is particularly interesting because, unlike many examples in this paper the count is not spread uniformly across signatures. That is, one might expect to prove this fact by showing that each signature $\Sigma$ has $C_n$ absolutely weakly increasing signed parking functions in $\mathcal{SPF}_n(\Sigma)$, but the following example shows that this is not the case.

**Example 5.7.** The set of absolutely weakly increasing signed parking functions of length 3 with signature $\{2\}$ consists of the four tuples

$$(1,-2,2),\ (1,-2,3),\ (2,-2,2),\ \text{and}\ (2,-2,3)$$

while the set of absolutely weakly increasing signed parking functions of length 3 with signature $\{3\}$ consists of the six tuples

$$(1,1,-3),\ (1,2,-3),\ (1,3,-3),\ (2,2,-2),\ (2,2,-3),\ \text{and}\ (2,3,-3).$$

Moreover, we are curious what an analog of signed parking functions that carry an action of the type $D$, rather than type $B$, Coxeter group would look like. Hence, we pose the following.

**Problem 5.8.** Give an analog of signed parking functions that carry an action of the type $D$ Coxeter group.

As final direction for further study, we recall the work of Celano et al. [2] who give a $q$-exponential generating function for inversions on parking functions. This was achieved via symmetric function theory and through a bijection to labeled rooted forests. A similar analysis for signed parking functions would be of interest.

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